A recurring, quality-gated S&P 500 pipeline

Dagster + Great Expectations + BigQuery – code and screenshots

Module 2 · 15 Oct 2026

Inside the sp500-gx-dagster repo

`start_dagster.sh` — used

One command: starts the web page + scheduler

`orchestration/` — used

assets.py: the steps · definitions.py: 2 jobs, 2 schedules

`data_quality/` — used

GX gates 0–4 · gx_suite.py helper · tables.py

`scripts/` — used

Parquet → BigQuery loader · SQL runner · Supabase copy

`team2_sql/` — used

Sharpe mart + 2 views (cut off at 7 Oct)

`runs/` — evidence

Saved gate results, v1 and v2 + 8 Oct evidence

`team_dbt_checks/` — off

6 dbt tests (no dbt profile set up yet)

`Dockerfile, compose` — not tested

Starter files for running in Docker

Starting Dagster and scheduling the job

Start everything

`start_dagster.sh`

```
conda activate dagster
export SP500_REPO=$(pwd)
# each step runs in the course env:
export RUN_PY="conda run -n elt python"
# run history + schedule state:
export DAGSTER_HOME=~/dagster_home_practice
dagster dev -m \
  orchestration.sp500_pipeline.definitions
```

Job + schedule

`orchestration/sp500_pipeline/definitions.py`

```
own_v2_job = define_asset_job(
    "own_v2_extract_and_load",
    selection=AssetSelection.groups("own_v2"))

own_v2_daily = ScheduleDefinition(
    job=own_v2_job,
    cron_schedule="30 6 * * 2-6",  # Tue-Sat
    execution_timezone="Asia/Singapore")
```

*"30 6 * * 2-6" = minute 30, hour 6, Tuesday to Saturday – after each US trading day.*

The three steps of the recurring job

Each step is a Dagster asset; deps= sets the order

`orchestration/sp500_pipeline/assets.py`

```
@asset(group_name="own_v2", retry_policy=RetryPolicy(max_retries=1, delay=300))
def extract_kenny_parquet(context):        # 1. run sp500_panel-2.ipynb ->
data/<today>/*.parquet
    cmd = "python -m jupyter nbconvert --execute sp500_panel-2.ipynb --output-dir runs ..."
    return done(sh(context, cmd, cwd=V2, env={"SP500_OUT": "data"}))

@asset(group_name="own_v2", deps=[extract_kenny_parquet])
def gx_gate0_parquet(context):             # 2. GX gate 0 on the files
    return done(sh(context, "python suite_0_parquet_extract.py", cwd=DQ))

@asset(group_name="own_v2", deps=[gx_gate0_parquet])
def load_own_v2_bq(context):               # 3. only runs if gate 0 passed
    return done(sh(context, "python scripts/load_parquet_to_bq.py"))
```

If a step fails, Dagster skips everything after it – bad data never reaches BigQuery.

GX gate 0: rules that must pass before loading

The rules

`data_quality/suite_0_parquet_extract.py`

```
between("n_rows", 600_000, 700_000)
between("tickers", 495, 510)
between("tickers_behind", 0, 5)
between("days_since_last_date", 0, 4)
between("duplicate_keys", 0, 0)
between("null_close", 0, 0)
between("bad_prices", 0, 0)

# bad_prices: open/close outside
# the day's high-low, or close <= 0
```

The gate

`data_quality/gx_suite.py`

```
results = checkpoint.run(
    batch_parameters={"dataframe": df})

for r in validation.results:
    print("PASS" if r.success
          else "FAIL", ...)

if not results.success and halt:
    raise Exception(
      "Data quality check failed
       - pipeline halted.")
```

between(…) is short for gxe.ExpectColumnValuesToBeBetween – one Great Expectations rule each.

Gate 3: the rule that caught 8 October

Price validity, measured in BigQuery

`data_quality/suite_3_michael_14_dimensions.py`

```
SELECT COUNTIF(close <= 0 OR adj_close <= 0
   OR high < low
   OR open > high OR open < low
   OR close > high OR close < low)
AS bad_prices

# critical: must be 0, or the pipeline halts
between("bad_prices", 0, 0)

run_suite(m, "michael14_critical_suite",
          critical, halt=True)
```

Critical

Must pass, or nothing downstream runs (duplicates, impossible prices)

Warning

Known, explained issues – reported every run, does not halt (gaps, zero-volume days)

v2 result

bad_prices = 20, all on 8 Oct → halted

Loading into BigQuery, and the 7 Oct cut-off

Parquet → my BigQuery, then verify

`scripts/load_parquet_to_bq.py`

```
TABLE = "friendly-autumn-509313-n3"
        ".sp500_v2.panel_daily"
LoadJobConfig(
  write_disposition="WRITE_TRUNCATE",
  time_partitioning=TimePartitioning(
      type_=MONTH, field="date"),
  clustering_fields=["ticker"])

if n_bq.n != len(df):
    raise SystemExit("Row count mismatch")
```

Sharpe mart cut-off

`team2_sql/01_build_mart_sharpe.sql`

```
SELECT ticker, date,
       daily_return
FROM proj2sp500
     .fact_dailyprice
WHERE daily_return
      IS NOT NULL
  AND date <= '2026-10-07'
  -- 8 Oct is intraday;
  -- remove after
  -- re-download
```

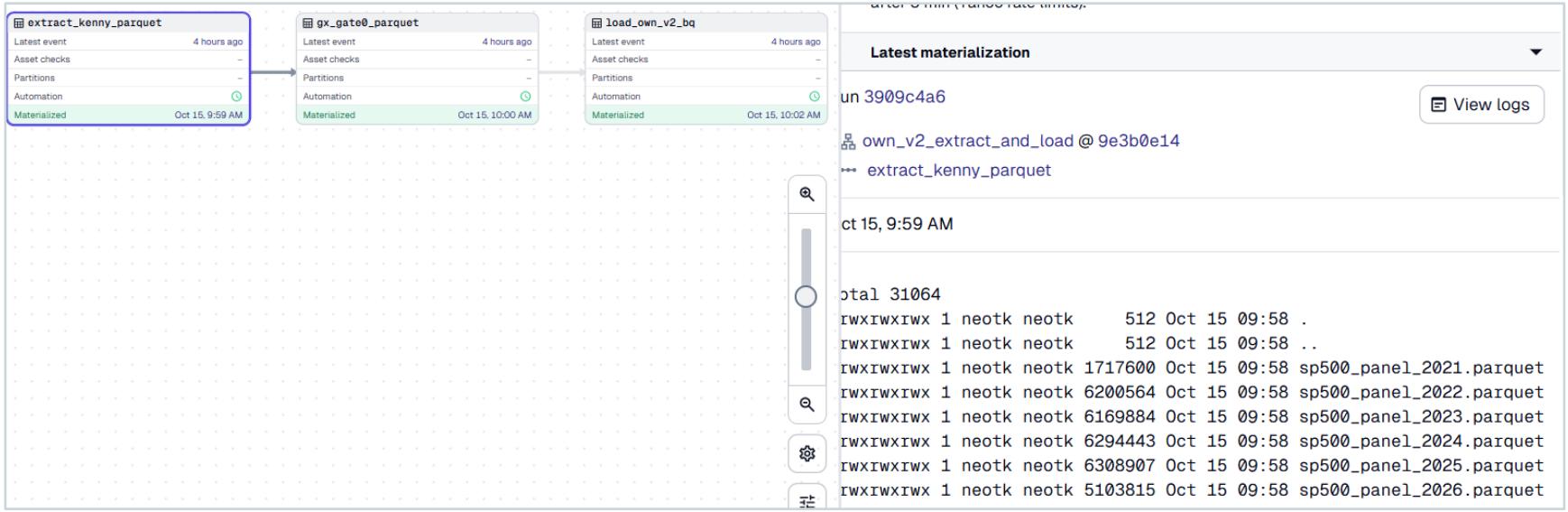
WRITE_TRUNCATE = full refresh each run; next step is an incremental MERGE on (ticker, date).

1 Two jobs in Dagster

Name	Schedules/sensors	Latest run
orchestration.sp500_pipeline.definitions		
own_v2_extract_and_load	At 06:30 AM GMT+8, Tuesday through Saturday	Oct 15, 9:40 AM
team_checks_and_marts	At 08:00 AM GMT+8, Tuesday through Saturday	Oct 14, 1:42 PM

- own_v2_extract_and_load: my recurring job, schedule on
- team_checks_and_marts: gates 1–4, schedule off
- Off on purpose until team data is refreshed

2 The job: three steps



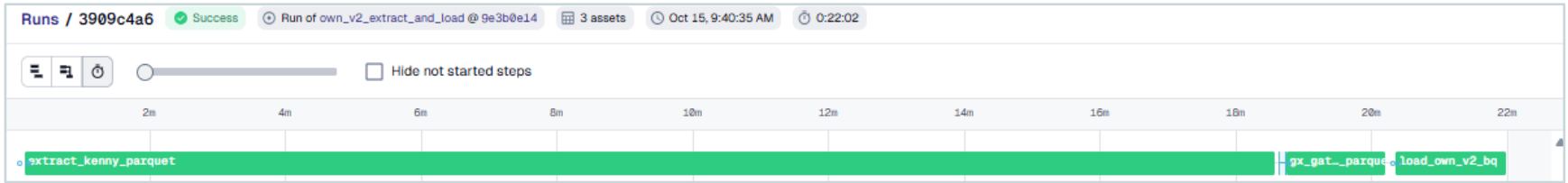
- Download: sp500_panel-2.ipynb (Wikipedia + Yahoo + SEC)
- GX gate 0 checks the files
- Load into my own BigQuery – never the team tables
- Right: the 6 Parquet files, one per year

3 The schedule

Latest tick	Schedule has never run
Next tick	Oct 16, 6:30:00 AM GMT+8
Target	own_v2_extract_and_load
Running	Reset schedule status
Partition set	None
Schedule	At 06:30 AM GMT+8, Tuesday through Saturday `(30 6 * * 2-6)`
Execution timezone	Asia/Singapore

- Tue–Sat at 6:30 am Singapore time
- Cron: 30 6 * * 2-6
- After the US close: complete prices
- First scheduled run: 16 Oct, 6:30 am

4 This morning's run



- Success, all 3 steps green
- 22 minutes end to end

- Most of it is the download (about 18 min)
- Retries after 5 min if Yahoo rate-limits

5 Gate 0: checks before loading

```
Checking 6 Parquet files in /mnt/c/Users/neote/projects/sp500_v2/data/
last date in files: 2026-10-14
                          0
n_files                   6
n_rows               622757
tickers                 503
tickers_behind            1
duplicate_keys            0
null_close                0
bad_prices                0
days_since_last_date      1
 PASS expect_column_values_to_be_between n_files 0 [completeness]
 PASS expect_column_values_to_be_between n_rows 0 [coverage]
 PASS expect_column_values_to_be_between tickers 0 [coverage]
 PASS expect_column_values_to_be_between tickers_behind 0 [completenes
 PASS expect_column_values_to_be_between days_since_last_date 0 [timel
 PASS expect_column_values_to_be_between duplicate_keys 0 [uniqueness]
 PASS expect_column_values_to_be_between null_close 0 [completeness]
 PASS expect_column_values_to_be_between bad_prices 0 [validity (intra
parquet_extract_suite: PASSED
```

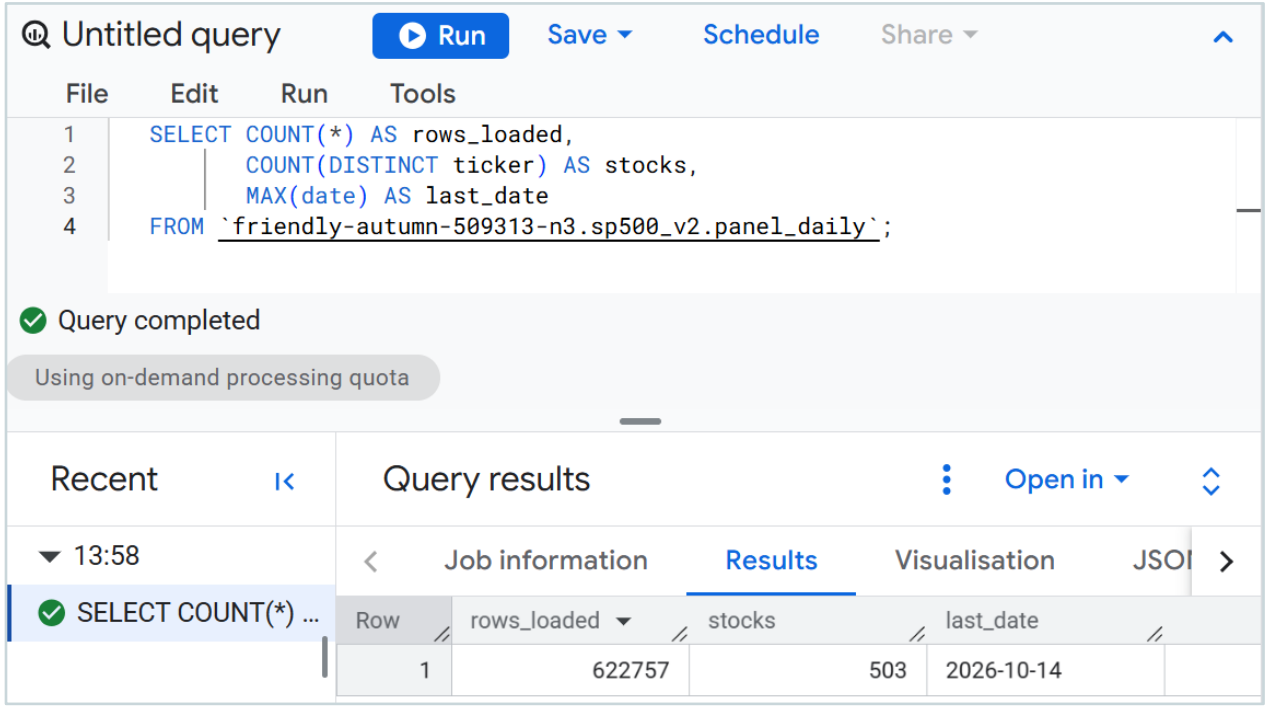
- 8 rules, all PASS
- Rows, stocks, duplicates, missing prices
- Impossible prices (open outside high–low)
- 1 stock a day behind: allowed (limit 5)

6 The load, with its own check



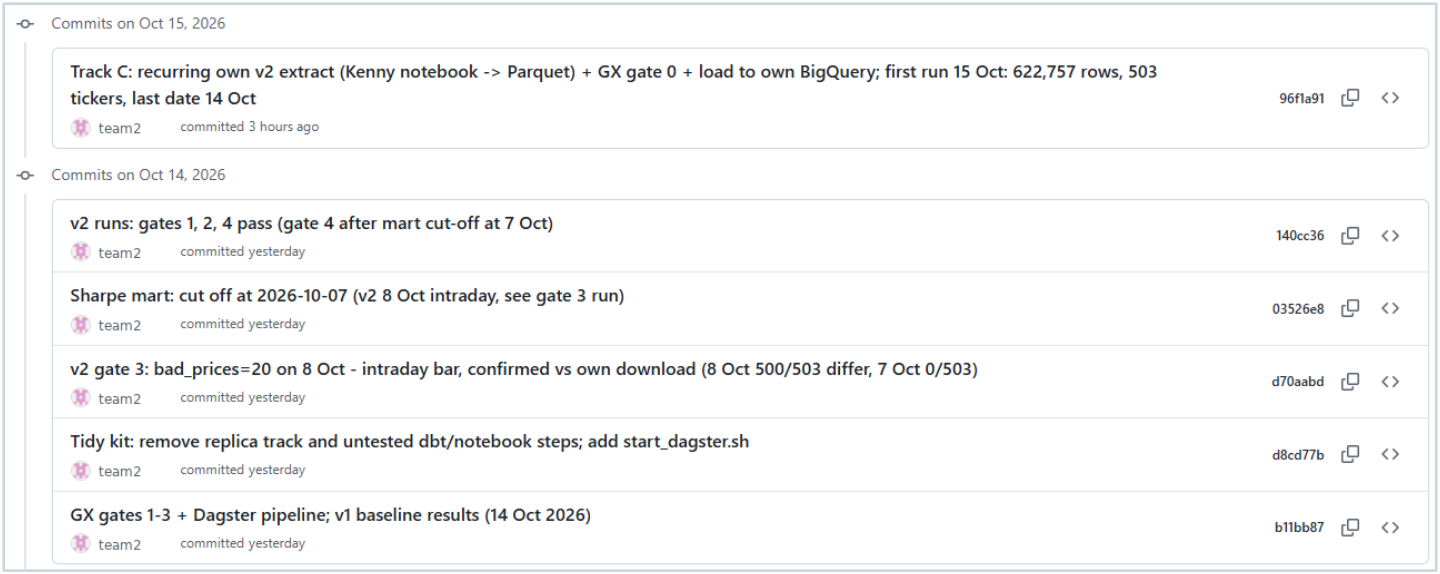
- 622,757 rows · 503 stocks
- Last complete day: 14 Oct
- Rows in BigQuery = rows in files

7 The result in BigQuery



- 622,757 · 503 · 2026-10-14
- Partitioned by month, clustered by ticker
- Parquet: 31 MB vs 183 MB CSV

8 Every step on GitHub



- 6 commits, 14–15 Oct
- Gates first, then gate 3 failure + proof
- Then the fix, then the recurring job

9 What the gates caught: 8 Oct

```
v2 gate 3 (suite 3) - 14 Oct 2026
CRITICAL FAIL: bad_prices = 20 (v1 = 0)
All 20 rows dated 2026-10-08 (last day in team v2), all "open outside high-low"
  open below low (16): URI EG UHS MCK PHM MA AIZ TPR DD TPL WRB GL ALL TKO GDDY MO
  open above high (4): CRL FTV PNW DGX

Cross-check vs my own download (sp500_v2.sp500_panel, ends 2026-10-13):
  2026-10-07: 503 tickers, 0 closes differ, avg diff 0.0%
  2026-10-08: 503 tickers, 500 closes differ, avg diff 1.437%

Conclusion: team v2 was downloaded while 8 Oct US session was still trading,
so 8 Oct is a partial intraday bar (not the final close) for all stocks.
Gate caught 20 where it is visibly impossible; the other 480 look valid but are wrong.
Action: last complete day for v2 = 2026-10-07. Gate rule NOT loosened.
```

- 20 impossible prices on 8 Oct
- 7 Oct: 0 of 503 closes differ
- 8 Oct: 500 of 503 differ
- Analysis ends 7 Oct; rule not loosened